

Lock-Based Protocols & Multiple Granularity — DBMS

Definition: Lock-Based Protocols control concurrent access to data items by requiring transactions to acquire **Locks** (Shared or Exclusive) before reading or writing data.

1. Detailed Technical Explanation

1. Lock Modes:

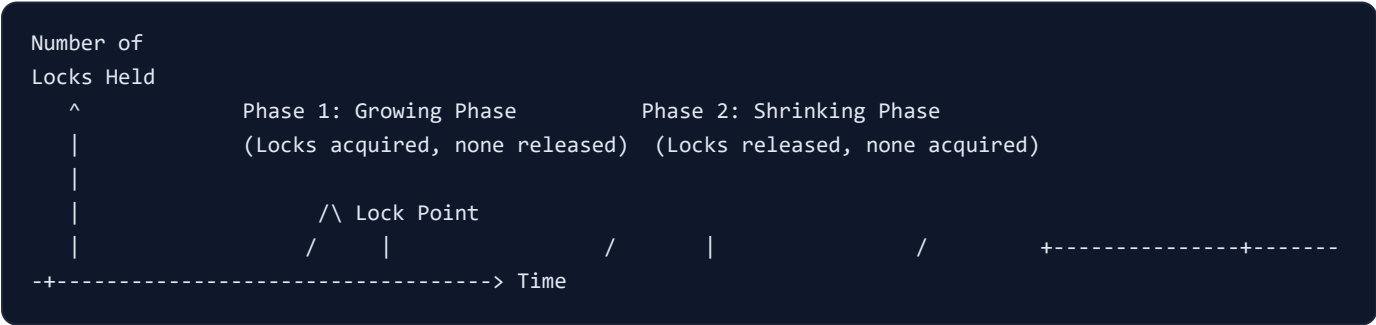
- 1. **Shared Lock (S-Lock):** Granted for **READ** operations. Multiple transactions can hold S-locks simultaneously on the same item.
- 2. **Exclusive Lock (X-Lock):** Granted for **WRITE** operations. Only one transaction can hold an X-lock; no other transaction can read or write.

Lock Compatibility Matrix:

Lock Requested	Shared (S)	Exclusive (X)
Shared (S)	Compatible (True)	Incompatible (False)
Exclusive (X)	Incompatible (False)	Incompatible (False)

2. Two-Phase Locking Protocol (2PL)

2PL guarantees conflict serializability by dividing a transaction's lock lifecycle into two distinct phases:



2PL Variants:

- 1. **Basic 2PL:** Transaction acquires locks as needed (Growing), releases locks (Shrinking). Can suffer from **Cascading Aborts**.
- 2. **Strict 2PL:** Transaction holds all **Exclusive (X) locks until COMMIT/ABORT**. Prevents cascading aborts (Strict schedules).
- 3. **Rigorous 2PL:** Transaction holds **ALL locks (Shared and Exclusive) until COMMIT/ABORT**.

3. Multiple Granularity Locking (MGL)

MGL allows data items of various sizes (Database -> File -> Page -> Record) to be locked in a tree hierarchy.

```
[ DATABASE ]
  |
[  FILE   ]
  |
[  PAGE   ]
  |
[ RECORD  ]
```

Intent Locks:

Before locking a fine-grained node (e.g., Record), a transaction must acquire an **Intent Lock** on its ancestor nodes:

- **IS (Intent Shared)**: Intent to lock explicit child nodes with Shared locks.
- **IX (Intent Exclusive)**: Intent to lock explicit child nodes with Exclusive locks.
- **SIX (Shared + Intent Exclusive)**: Explicit Shared lock on current subtree plus Intent Exclusive on lower nodes.

4. Core Concepts & Memory Keywords

- **Lock Point**: Point in time where a transaction acquires its final lock in 2PL.
- **Cascading Rollback**: One transaction aborting causes multiple dependent transactions to abort.
- **Intent Locking**: Top-down locking mechanism in granularity trees.

5. Must-Write Points for Exams

- Basic 2PL guarantees **conflict serializability**, but does NOT prevent **deadlocks**.
- Strict 2PL prevents cascading aborts by holding Exclusive locks until the transaction commits.
- Multiple granularity locking improves concurrency by allowing fine-grained locks without searching entire trees.

6. Quick Recall Flow

Growing Phase (Acquire Locks) -> Lock Point -> Shrinking Phase (Release Locks) -> Hold X-Locks to Commit (Strict 2PL)